

EduGenie – Project Demonstration

1. Project Introduction

EduGenie is a Google Gemini powered learning assistant developed to support students in learning and understanding different topics.

2. Problem Addressed

Students may face difficulties when understanding complex concepts, revising information, practicing questions and deciding what to study next.

EduGenie combines several AI-powered learning features in one application.

3. Technologies Used

- Python
- FastAPI
- Uvicorn
- HTML
- CSS
- Jinja2
- Google Gemini API
- GitHub

4. Main Features

Question and Answer

Provides AI-generated answers to student questions.

Simplified Explanation

Explains complex topics in simpler language.

Quiz Generation

Generates quiz questions for practice.

Text Summarization

Creates concise summaries from provided text.

Learning Path Recommendation

Provides suggested learning steps for a selected topic.

5. Demonstration Flow

The project demonstration follows these steps:

1. Introducing the EduGenie project.
2. Explain the problem statement.
3. Explain the project objectives.
4. Introduce the technologies used.
5. Explain the system architecture.
6. Explain the main modules.
7. Demonstrate the major learning features.
8. Explain the testing process.
9. Discuss future enhancements.

6. Expected Demonstration Outcome

The demonstration should show how EduGenie can be used as an AI-powered learning assistant and how its different features support students.

7. Future Enhancements

Future versions may include:

- Voice interaction
- Multilingual support
- Mobile application
- Student progress tracking
- Gamification
- Image and PDF based learning
- Learning Management System integration

8. Conclusion

EduGenie demonstrates how generative AI can be integrated with a web application to provide multiple educational features through a single learning assistant.